Geometry

(Beginner)

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Geometry Problems

- Typically involves a lot of tiny details and long codes. Takes a good amount of time to get AC.
- Ideally you should start coding them before the 5th hour starts (or even earlier).
- Be careful when dealing with fractions.

Important Advice

- Don't use floating point if you don't have to. Many problems can be solved by only using integers. Work things out on paper.
- If you still have to use them, try to delay them as much as possible. Operations like: division, sqrt and trigonometric functions are the biggest source of precision loss.
- Make sure you spot all the cases on paper. Try to write smart code that handles many at once.

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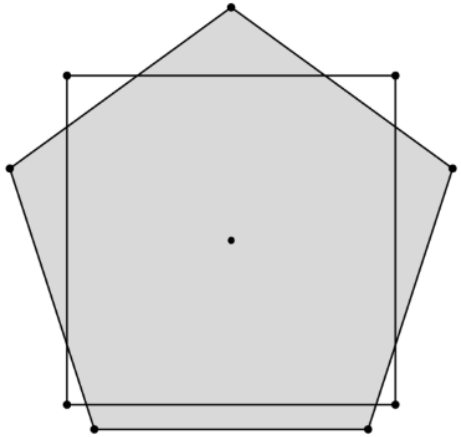
Rotating Painter

- Two regular convex polygons are pinned at their center. The top polygon can be rotated freely.



- You can paint only the visible parts of the bottom polygon.
- Print the total area of the bottom polygon that you can color.
- testcases $\leq 10^5$, each polygon has ≤ 1000 vertices, TL: 1 sec.

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Solution

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Points and Vectors

- We treat points and vectors with the same data structure.

```
template<class T> struct Point {
  typedef Point P;
  T x, y;
  explicit Point (T _x=0, T _y=0) : x(_x) , y(_y) {}

  friend ostream& operator<<(ostream& os, P p) {
    return os<<"("<< p.x << "," << p.y << ")";
  }

  P operator+(P p) const { return P(x + p.x, y + p.y); }
  P operator-(P p) const { return P(x - p.x, y - p.y); }
  P operator*(T d) const { return P(x * d, y * d); }
  P operator/(T d) const { return P(x / d, y / d); }
```

Points and Vectors

The Dot and Cross Products

- Let's understand what these two operators do.

```

  T dot(P p) const { return x * p.x + y * p.y; }
  T cross(P p) const { return x * p.y - y * p.x; }
  T cross(P a, P b) const { return (a - *this).cross(b - *this); }

```

Points and Vectors

The Dot Product

- Can be used to calculate the angle between two vectors.
- Can be used to calculate the projection length.
- It can be used to check if a vector is perpendicular to another.

Points and Vectors

The Cross Product

- Can be used to find the angle between two vectors.
- Can be used to calculate the area* of a triangle/parallelogram formed by the vectors.
- Can be used to find the orientation of two vectors.
- Can be used to detect co-linear points.

Points and Vectors

– Finally some more useful operations.

```
  T dist2() const { return x * x + y * y; }
  double dist() const { return sqrt((double)dist2()); }

  // angle to x-axis in interval [-pi, pi]
  double angle() const { return atan2(y, x); }
  // makes dist() = 1
  P unit() const { return *this/dist(); }
  // rotate by +90 degree
  P perp() const { return P(-y, x); }
  P normal() const { return perp().unit(); }
  //rotate 'a' radians ccw around (0,0)
  P rotate(double a) const {
    return P(x * cos(a) - y * sin(a),
             x * sin(a) + y * cos(a));
  }

  bool operator<(P p) const { return tie(x, y) < tie(p.x, p.y); }
  bool operator==(P p) const { return tie(x, y) == tie(p.x, p.y); }
};
```

Lines

- The standard method of dealing with lines is to keep them in the $ax + by = c$ form.
- Parallel lines checking, Equal lines checking, Finding intersection points, Finding perpendicular line, Finding bisector of two points.

Segments

- We store segments by their endpoints. And use line equations if needed.
- For checking if a point is on a segment:

```
bool inDisk(pt a, pt b, pt p) {
    return dot(a-p, b-p) <= 0;
}
```

```
bool onSegment(pt a, pt b, pt p) {
    return orient(a,b,p) == 0 && inDisk(a,b,p);
}
```

Segment Segment Intersection

- To check whether two segments intersect or not is very common subroutine and we explain it here.

```
template<class P> vector<P> segInter(P a, P b, P c, P d) {
  auto oa = c.cross(d, a), ob = c.cross(d, b),
       oc = a.cross(b, c), od = a.cross(b, d);
  // Checks if intersection is single non-endpoint point.
  if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
    return {(a * ob - b * oa) / (ob - oa)};

  set<P> s;
  if (onSegment(c, d, a)) s.insert(a);
  if (onSegment(c, d, b)) s.insert(b);
  if (onSegment(a, b, c)) s.insert(c);
  if (onSegment(a, b, d)) s.insert(d);
  return {all(s)};
}
```

Area of a polygon

- How to find the area of a polygon? Given all the points that define it.

Point Inside Polygon

- There are at least 3 well known approaches to this.

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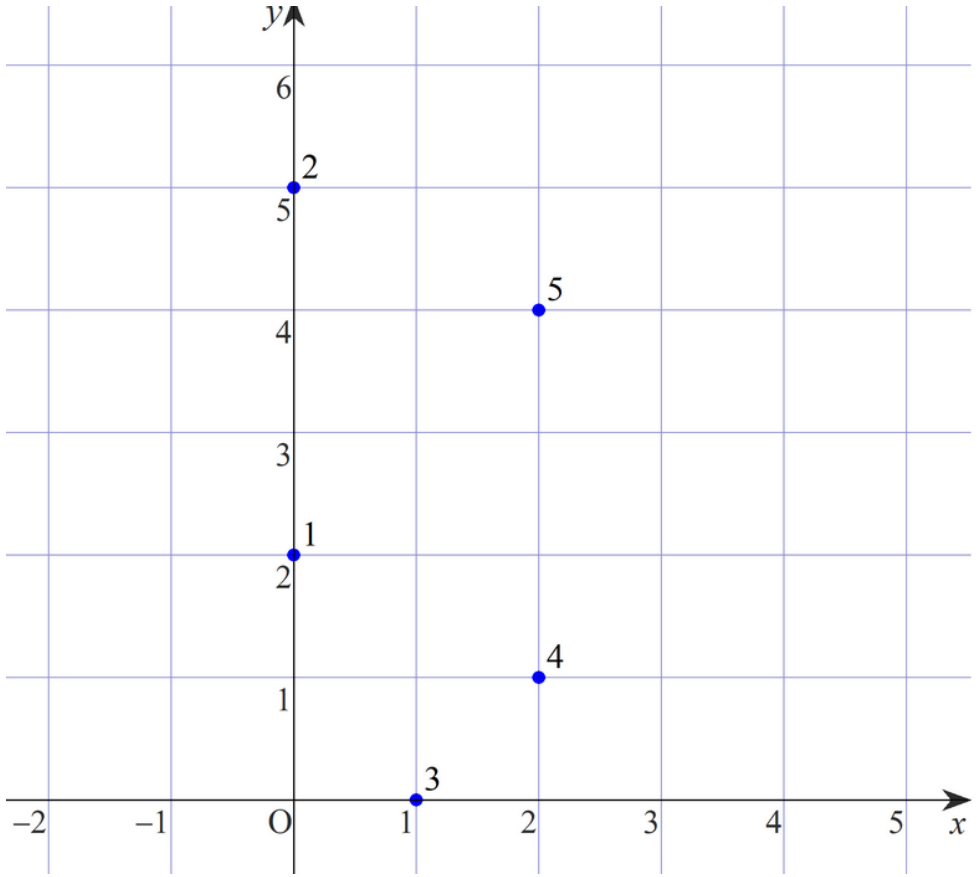
- The first one uses the area.
- The second method is known as ray shooting.
- The third approach is for convex polygons.

Geometry Primitives Contest

- You can practice your template usage skills in this contest: https://codeforces.com/gym/100168

ABC 418 - E

- Given N points in the 2D plane, count how many combinations of 4 points form a trapezium. $N \leq 2000$.



ABC 418 - E

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Lattice Points

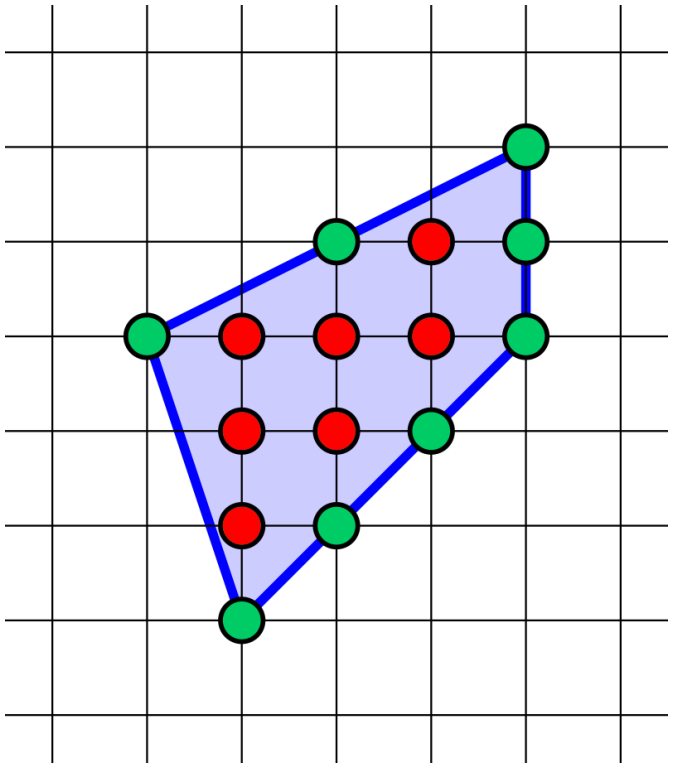
- On the segment (x_1, y_1) to (x_2, y_2) how many lattice points exist?

Lattice Points

- On the segment (x_1, y_1) to (x_2, y_2) how many lattice points exist?
- The answer is $\gcd(|x_1 - x_2|, |y_1 - y_2|) + 1$

Pick's Theorem

- $A = I + \frac{B}{2} - 1$



(source: wikipedia)

2016-2017 ACM-ICPC Asia-Bangkok Regional Contest - D

- Given a segment (x_1, y_1), (x_2, y_2).
- We wish to find K points (x, y) that satisfy the following:
- The segments from (x, y) to either of the given points have no lattice points on them.
- The triangle created by the 3 points have no lattice points inside.
- The triangle has positive area.
- testcases ≤ 1000, $|\text{co-ordinates}| \leq 10^9$, $\sum K \leq 20000$.

2016-2017 ACM-ICPC Asia-Bangkok Regional Contest - D

Solution

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SWERC 2024 - E

- We need to build a simple regular polygon with at most $3N$ vertices, where N vertices are already given.
- The polygon can not have any lattice points inside.
- $N \leq 1000$.

SWERC 2024 - E

Solution

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Codechef POLYCON

- A polygon is said to have K intersections if exactly K unordered pairs of its edges properly intersect. A pair of edges of a polygon is said to properly intersect if they are not parallel to each other, and intersect at a point that is not an endpoint of either edge.
- For each K from 0 to 10,000 print a polygon with upto 150 vertices with exactly K intersections.

Codechef POLYCON

Solution

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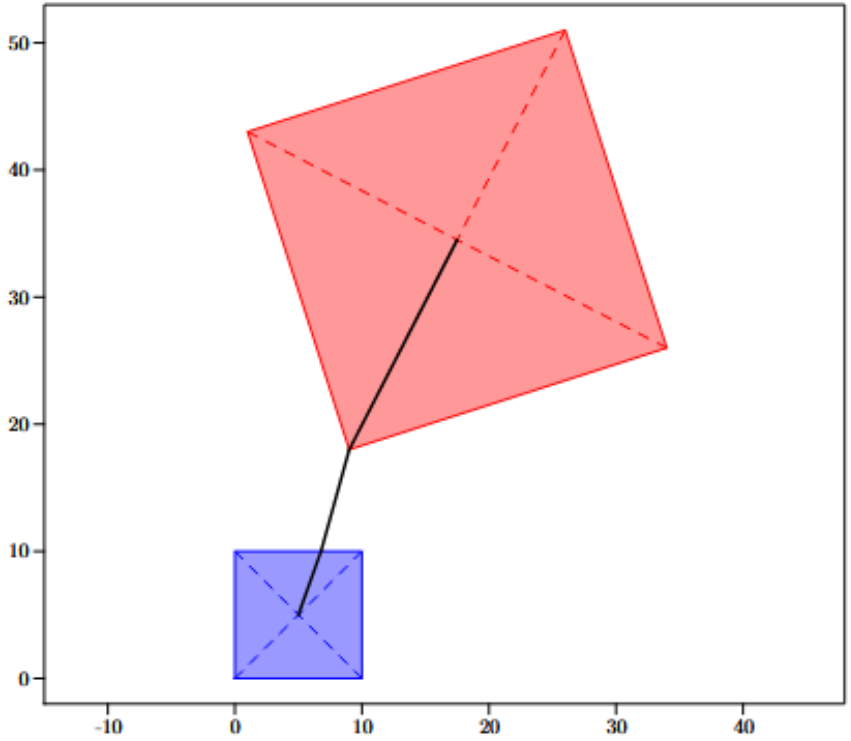
Ternary Search

- Super useful technique to solve geometry problems. Often you can just guess the ideas.

ICPC WF 2023 - D

- There are two pyramids, defined by one of their edges, and the height h at the center.



- Task is to go from one peak to another peak. Find the minimum distance to walk.

ICPC WF 2023 - D

Solution

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2019-20 Winter Petrozavodsk, Day 8 - F

- I have drawn n dots on a plane and m straight segments connecting these dots. No segment goes through dots other than its endpoints, and no two segments intersect in any point other than their common endpoint. Also, if you start in one dot and move only by segments, you can go to any other dot. All n dots are in distinct positions.
- Your task is to check if each face of this graph, including the outer face, is a triangle. Face is a triangle if and only if it is bounded by exactly 3 edges. If face is on both sides of some edge, this edge is counted twice
- $n \leq 10^5$, $m \leq 3 \times 10^5$, TL: 2 sec

2019-20 Winter Petrozavodsk, Day 8 - F

Solution

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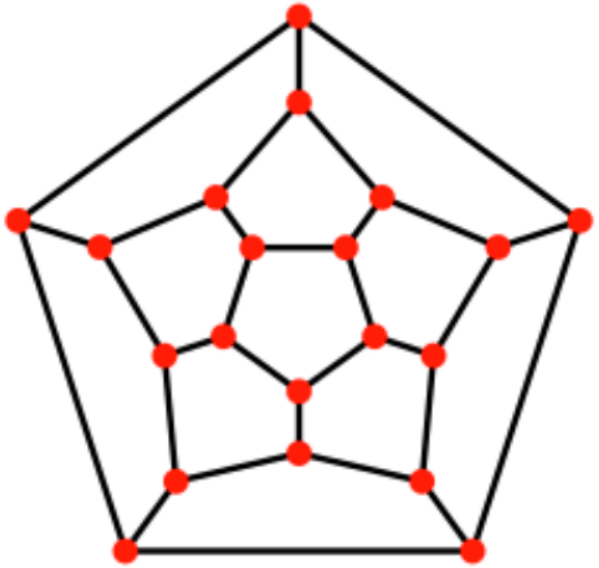
Euler's Formula for Planar Graphs

- The formula states: $V - E + F = 2$



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2019-20 Winter Petrozavodsk, Day 8 - F

Solution

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Extra Problem

- Given a convex polygon defined by n points, find the total area of all interior triangles.

– Good luck!